

Math 290

Practice Quiz 2 — Version D

(extra practice, Sections 6–11)

1. Short response.

- (a) Evaluate the following proposed theorem and proof.

Theorem: Let $x, y \in \mathbb{Z}$. If xy is even, then x is even or y is even.

Proof. Let $x, y \in \mathbb{Z}$. Assume that xy is even. We consider two cases.

Case 1: Assume x is even. Then x is even or y is even, as desired.

Case 2: Assume y is even. Then, again, x is even or y is even.

In every case, x is even or y is even. □

- A. The theorem is true and the proof is correct.
 - B. The theorem is false.
 - C. The theorem is true, but the proof is incorrect because its cases do not cover the possibility that x and y are both odd.
 - D. The theorem is true, but the proof is incorrect because it should have been done by contradiction.
 - E. The theorem is true, but the proof is incorrect because Case 2 is the same as Case 1 and should be omitted.
- (b) True or false: If $x \in \mathbb{R}$ and $x < 0$, then $|-x| = x$.
- (c) True or false: The negation of “For all $x \in \mathbb{Z}$, if x is even, then x^2 is even.” is “There exists $x \in \mathbb{Z}$ such that x is even and x^2 is odd.”

2. Let $a \in \mathbb{Z}$.

- (a) Prove that if $4 \nmid a^2$, then a is odd.
- (b) State the converse of the implication in part (a), and then prove or disprove it.

3. Prove that $\sqrt[3]{2}$ is irrational.

4. Fix

$$A = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : 4 \mid (x - y)\},$$

$$B = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x \text{ and } y \text{ have the same parity}\}.$$

Prove that $A \subseteq B$. Then prove or disprove: $B \subseteq A$.